

Lily Zhang

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CAREER STATEMENT

I'm a Technical Project Lead in AI & ML with 8 years of experience, and I like turning research into products people use. At Latitude AI, I build LLMs, multimodal LLMs, and AI agents for physical AI. My focus: agentic post-training and AI training data, incl. RL environments. Open-sourced: [SofaGenius](#) (Anthropic hackathon finalist, top 15/13k), [Frontend Slides](#) (25K+ GitHub stars), and the [Deep-Seek](#) research agent (500+ GitHub stars). Recent work: [AURA](#), hierarchical failure taxonomy on Harbor / Terminal-Bench; [LosslessBench](#); [C-Guard](#) (COLM 2026 ER).

HIGHLIGHTS

- **Benchmark Products:** Shipped [LosslessBench](#) (5 domains from real OpenRouter task traffic, incl. Terminal-Bench coding) and [SuperGeneral](#) (tool composition + creation eval across 5 frontier models).
- **AURA: Hierarchical Failure Taxonomy for Coding Agents:** Categorized coding-agent failures with executable detectors, 84% verifier failures; extracted the main modes: premature completion, unaddressed errors, command loops, wasted commands, missing environment, JSON parse warnings, context pressure, each routed to SFT or reward penalty.
- **Post-Training with Harbor RL Environment Rollouts:** Built a multi-turn RL environment on Terminal-Bench 2; rollout-driven reward shaping from the failure taxonomy; decontaminated 140K SFT trajectories.
- **Product Win:** Post-trained model (SFT + GRPO) in production at Alibaba Cloud, achieved reliable GPU demand forecasting with 91% of generated warnings adopted by ops and +16% demand fulfillment.
- **Eval-Driven Buy-In:** Aligned with leadership for the org's first VLM workstream with a 37-page proposal systematically evaluating state-of-the-art vision-language models, w a diffusion-generated long-tail safety validation set and temporal-causal reasoning tests.
- **Expert Network Operations:** Sourced and evaluated 10+ annotation and synthetic-data vendors (inc. Scale AI) at \$1M contract scale, running pilot-batch POCs before major purchases.

WORK EXPERIENCE

Latitude AI · Technical Lead, AI & ML · Feb 2024 – Present

- **Aqua, Agentic Label QA:** Defined the platform end to end: label QA as tool-use, with playbook-carrying tools and bring-your-own label sources; designed a two-phase architecture (offline tool discovery against an expensive-LLM baseline, frozen tools served online as a deterministic MCP server) to work within sandbox permissions.
- **VLM Adoption:** Introduced VLMs to the organization through systematic evaluation of state-of-the-art models, then shipped the production data-curation agent now used by 76+ engineers org-wide, the foundation for hybrid search and downstream applications.
- **GPU Allocation at Scale:** Owned 1,000+ GPU allocation across competing teams with quota tables, a priority queue, and automatic retries; delivered a 50K-model training campaign on Ray inside a fixed two-month window with no added capacity, and reallocated the team off two workstreams mid-quarter.
- **Training Crisis Recovery:** Unblocked a stalled training run in ~6 hours across time zones with root-cause debugging (a missing cache overloading HPC file counts) and a clean handoff doc; mentored 3 ML engineers to +8.1% GEO F1 over baseline within quarter.

Ford Greenfield Labs · Technical Project Lead, AI & ML · Jun 2022 – Jan 2024

- **Continuous Learning Data Engine:** Proposed and shipped the 3D auto-annotation engine behind Ford BlueCruise offline perception as its #1 code contributor; the first 15K sequences saved \$1.1M in labeling, and the pipeline is still running 3+ years later, powering 492K vehicles.

- **Research-to-Product Transition w Adoption Metrics:** Transitioned the team from R&D to applied delivery and tracked the shift with quantifiable measures: unit-test coverage for code quality, PR velocity from near zero to 2+ PRs per engineer per week.
- **Vendors & Data Contracts:** Sourced and evaluated 10+ annotation and synthetic-data vendors with pilot-batch POCs; ran a 10+ engineer data program without authority.

Ford Greenfield Labs · Sr. Research Scientist · Jun 2019 – May 2022

- **Pro Trailer Hitch Assist:** Turned synthetic-data research into a shipped feature on F-150, Lightning, and Super Duty (up to 50% real-data replacement); Henry Ford Technology Award, top <0.1% of ~177K employees.

PRODUCTS

- **LosslessBench** (2026): measures what inference acceleration costs in model behavior; GLM 5.2 frontend quality dropped 5.6 points under vendor acceleration while model-owned acceleration lost 0.3. [Blogpost](#) · [Dataset](#)
- **SuperGeneral** (2026): compositional tool environments across 4 domains; no frontier model achieved tool composition + creation across all. [Project Page](#) · [Code](#)
- **SofaGenius** (2026): agentic post-training platform: automated GRPO runs on an 80B MoE with cost-visible launches and convergence anomaly detection. [Project Page](#) · [Video](#)

PUBLICATIONS

- **C-Guard: Constitution AI for Data-Efficient RL Alignment** (COLM 2026 ER): post-trained a 4B guard model with GRPO; over-refusal 22.4% → 12.8%, balanced accuracy 0.876 → 0.921. [Paper](#) · [Blogpost](#) · [Code](#) · [Dataset](#)
- **Eureka: LLM-Driven Demand Prediction** (NeurIPS-W 2025, DASFAA 2026 Oral). [Paper](#) · [Poster](#)
- 8 earlier vision/data papers (CVPR, ICCV, WACV) and 7 granted patents on synthetic data generation and curation, cited by NVIDIA, GM, and Salesforce. [scholar](#)

LEADERSHIP & SERVICE

- **NVIDIA GTC Conference:** Keynote on AI training data generation @ GTC 23; Panelist at AI/ML infra @ GTC 26.
- **Industry Advisory Board:** Board Advisor for Texas A&M University and University of Delaware.
- **Research Advisor, AliCloud (2025):** Led the research direction for Eureka, AI infra agent post-training.

SKILLS

- **Product Management:** portfolio roadmap and strategy, product lifecycle from ideation and task taxonomy design to pilot, 0-to-1 product definition, stakeholder and executive communication, translating market signals into strategy.
- **Evaluation:** benchmark design (SWE-Bench-class coding benchmarks, Terminal-Bench lineage), failure taxonomy, data decontamination, reproducibility governance, dataset-bias detection, VLM-as-judge, human-in-the-loop eval.
- **Post-training:** SFT, GRPO, reward design, RL environments (Harbor / Terminal-Bench / τ^2 -bench)
- **Technical:** Python, SQL, PyTorch, Ray, Dagster, Docker, MCP servers, dashboards

EDUCATION

- Stanford University · Graduate Program, AI, School of Engineering (2022–2024)
- Penn State University · B.S. Computer Science (2013–2017)